

Modality-Driven Search with Holistic Trace Judging for ARC-AGI-2

72.9% on ARC Prize Verified — built on three frontier models, not one

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ARC-AGI-2 — a brief history, and two example problems

History of ARC-AGI

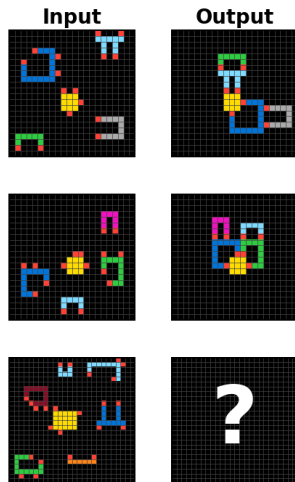
- **2019** — ARC-AGI-1 introduced (Chollet)
- **2020–2023** — Program synthesis era (Icecube et al.); top scores creep up to $\sim 20\%$ on ARC-AGI-1
- **Dec 2024** — OpenAI o3 saturates ARC-AGI-1: $\sim 76\%$ at \$200/task
- **2025** — ARC-AGI-2 launches: harder + cost-weighted; top frontier models $\sim 50\%$
- **Dec 2025** — **This work submitted: 72.9% on ARC-AGI-2**, first verified $>70\%$
- **2026** — Frontier models leapfrog to $\sim 80\%+$ on ARC-AGI-2 (GPT-5.5, Gemini 3 Deep Think); ARC-AGI-3 launched (top scores still $<1\%$)

ARC-AGI Example 1



e3721c99

ARC-AGI Example 2



cbebaa4b

What is ARC-AGI-2? — format and evaluation

What the model sees (JSON):

```
{
  "train": [
    {"input": [[0,1,0],[1,0,1],[0,1,0]],
     "output": [[1,0,1],[0,1,0],[1,0,1]]},
    {"input": [...], "output": [...]},
    ...
  ],
  "test": [
    {"input": [[1,0,1],[0,1,0],[1,0,1]]}
  ]
}
```

What the model returns:

```
[[0,1,0],[1,0,1],[0,1,0]]
```

- **pass@2**: the model is allowed *two guesses* per test input — it scores if either matches the ground truth exactly
- **Public eval** for development; the **private eval** is the official scoring — run end-to-end by the ARC Prize foundation on tasks the developer never sees
- **Cost per task (USD)** is graded alongside accuracy — efficiency is a first-class metric

The headline result

System	ARC-AGI-2	Cost/Task
Human Panel	100.0%	\$17.00
This work (semi-private)	72.9%	\$38.99
This work (public eval)	76.1%	\$19.69
GPT-5.2 Pro (High)	54.2%	\$15.72
Gemini 3 Pro (Refine.)	54.0%	\$30.57
Opus 4.5 (Thinking, 64K)	37.6%	\$2.40

+18.7 points over the strongest standalone frontier model

The real problem is selection, not generation

- LLMs produce fluent, internally coherent traces — and are still **confidently wrong**
- On hard tasks, models **cluster around the same wrong interpretation**
- $\Rightarrow$ majority voting / self-consistency *amplifies* the error
- The hard problem isn't generating a correct candidate
- It's **recognizing** one when most others disagree

Core idea: modalities as search operators

TEXT

Model reads the grid as numbers / strings

IMAGE

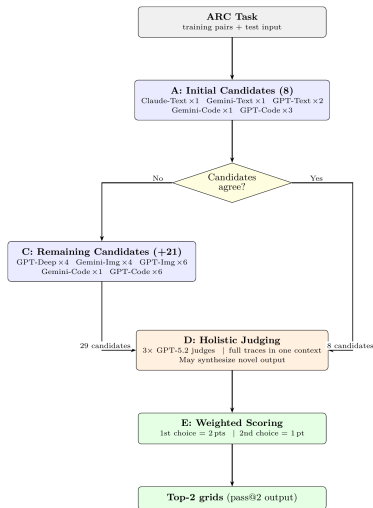
Model sees a rendered PNG of the grid

CODE

Model writes Python to transform the grid

- Same task, three representations → three different hypothesis distributions
- Each modality solves tasks the others miss
- Generate **independently** — don't pool prompts, don't share scaffolding
- Diversity at the *representation* level, not just temperature

Pipeline overview



Generate broadly → judge holistically → vote weighted

- Up to 29 candidates across Text / Image / Code, 3 frontier models
- 3 parallel judges read *all* traces in one long-context prompt
- Adaptive early-stop: 37/167 stopped after first probe; **36 correct (97%)**

Candidate generation in practice

- **3 models:** GPT-5.2 (x-high), Gemini 3 Preview (high), Claude Opus 4.5 (long-context)
- **3 modality families:** Text, Image, Code
- **29 candidates per task** across families $\times$ configurations
- **Deliberately minimal prompts** in every lane (*see slide 10*)
- Code candidates use a sandbox REPL for iterative debugging

Holistic judging vs. majority voting

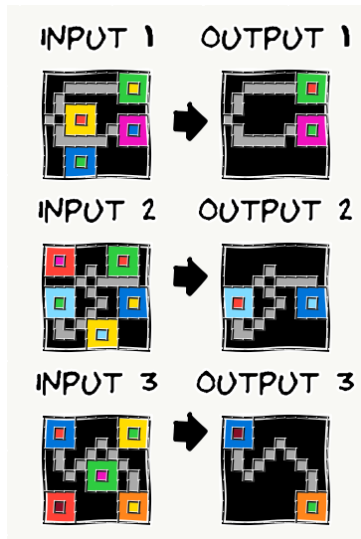
- Concatenate all 29 reasoning traces into one long-context judge prompt
- Each judge picks top-2 candidates *and* explains why other clusters are wrong
- Weighted vote across 3 judges $\rightarrow$ pass@2 guesses

Net uplift vs. majority-vote baseline: +7 instances

All 7 are minority recoveries — the correct answer wasn't the most common candidate.

Plus +1 **synthesis** instance (21897d95:2): zero correct candidates — judge recombined partial insights into a novel correct output.

End-to-end walkthrough: d35bdbdc



From 29 candidates → 2 guesses

Modality complementarity



- Pairwise non-overlap ($n=130$): Code solves 18 that Text misses; Image solves 13 that Text misses
- Exclusive solves: **2 Text-only, 6 Image-only, 7 Code-only**
- Modalities are **not redundant copies** of the same reasoning process
- Candidate-oracle accuracy: **86.2%** (a correct candidate exists for 144/167)

Negative results worth knowing

Prescriptive prompting degrades performance

Structured templates, CoT scaffolding, domain heuristics — all hurt on the hardest tasks.

Mechanism: a *compliance tax on reasoning*.

Iterative refinement reduces diversity

Refining candidates against each other collapses them onto the same wrong cluster — the exact failure mode we're trying to escape.

⇒ **Final system: deliberately minimal prompts + independent generation**

① **Modalities are search operators.**

Text, image, code activate different hypothesis distributions — exploit that.

② **Holistic judging > majority voting.**

On hard tasks the truth is a minority hypothesis; let a long-context judge read full traces.

③ **Less prompt structure, more representational diversity.**

Prescriptive scaffolding suppresses the creative leaps these tasks reward.

Backup: cost breakdown

- **Public-eval cost is the representative number: \$19.69/task**
- Semi-private \$38.99/task inflated by GPT-5.2 API instability (84% failure rate, 2,216/14,106 attempts succeeded)
- Tool-integrated code generation is the largest line item
- ZDR mode used in semi-private verification disables tool calls → both lower accuracy and altered cost profile
- ARC-AGI-1 on the same system: 94.5% at \$11.40/task — easier tasks early-exit cheaply

Backup: failure decomposition (167 instances)

- **128 solved (76.6%)**
- 21 generation failures (no correct candidate in pool)
- 1 early-stop failure (dbff022c:1 — extreme groupthink on a legend ambiguity)
- 17 **selection** failures (correct candidate exists, judge missed it)
- 1 synthesis success (21897d95:2)

Most future headroom is in selection, not generation.

Backup: why three judges?

- Single judge is biased toward its own prior
- 3 parallel judges + weighted scoring (2 pts first, 1 pt second) hedges *within* the judging step
- Pass@2 format lets the final output also hedge across two interpretations when judges disagree